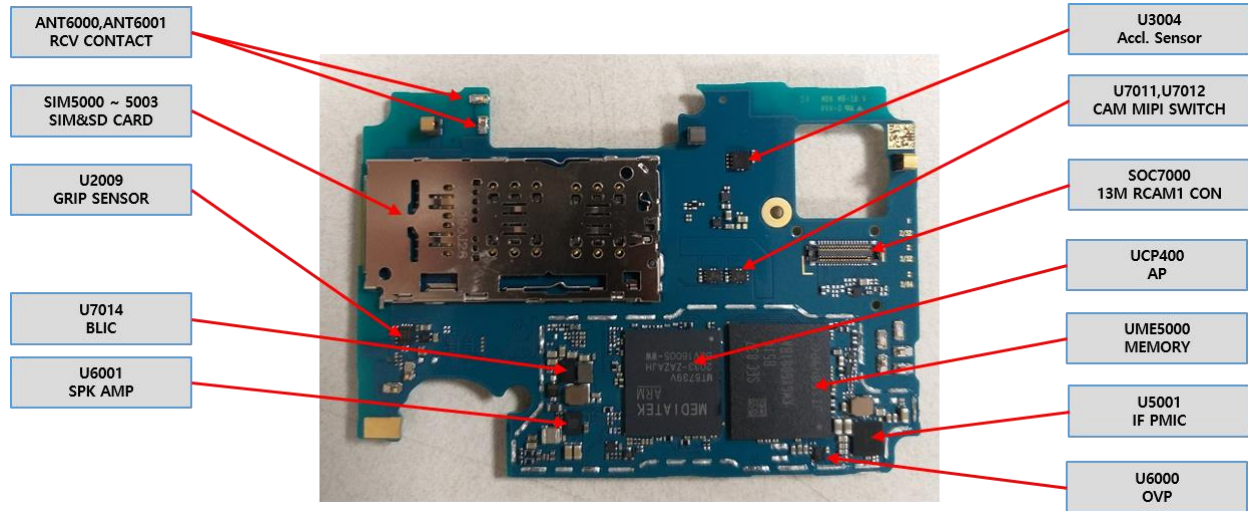


8. Level 3 Repair

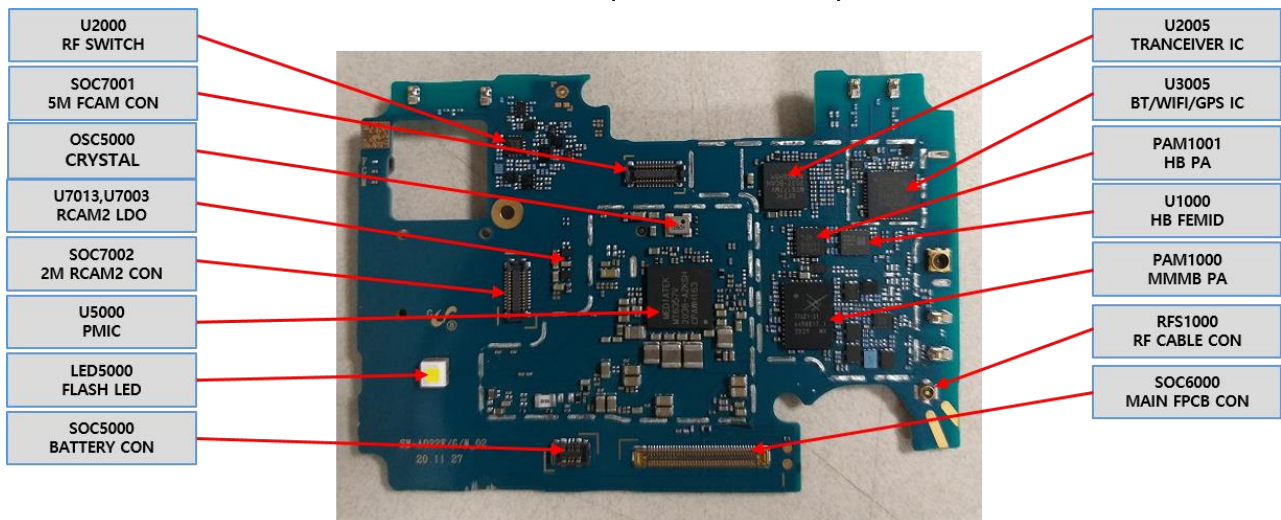
8-1. Components Layout

MAIN PBA (TOP SIDE)

Helpful-Man

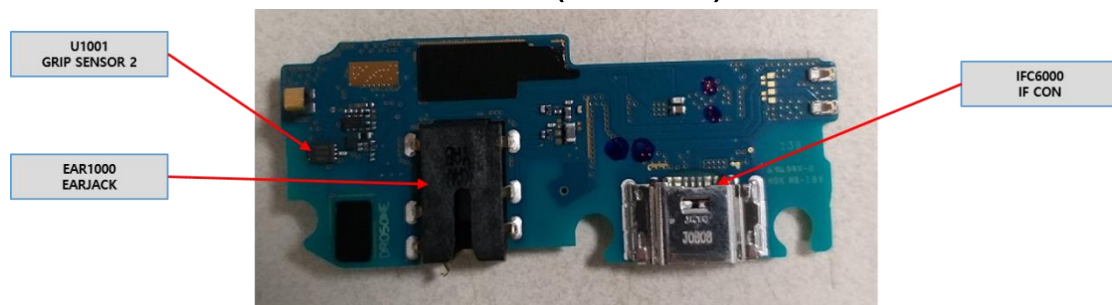


MAIN PBA (BOTTOM SIDE)

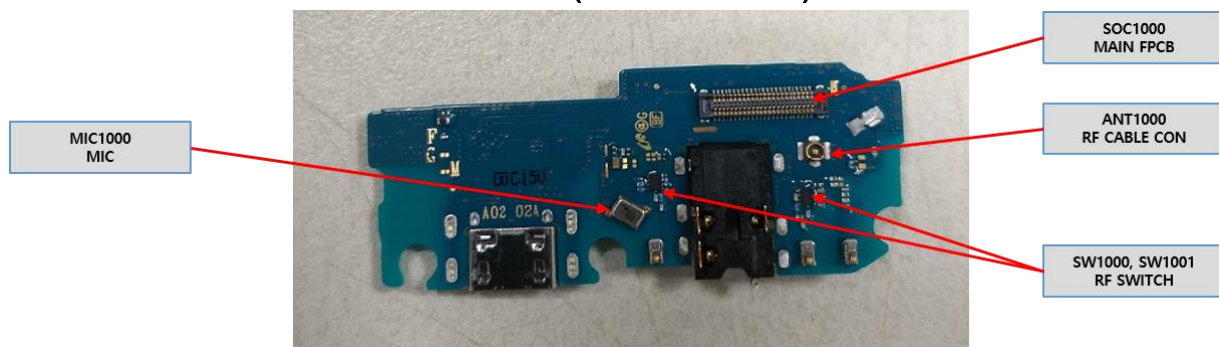


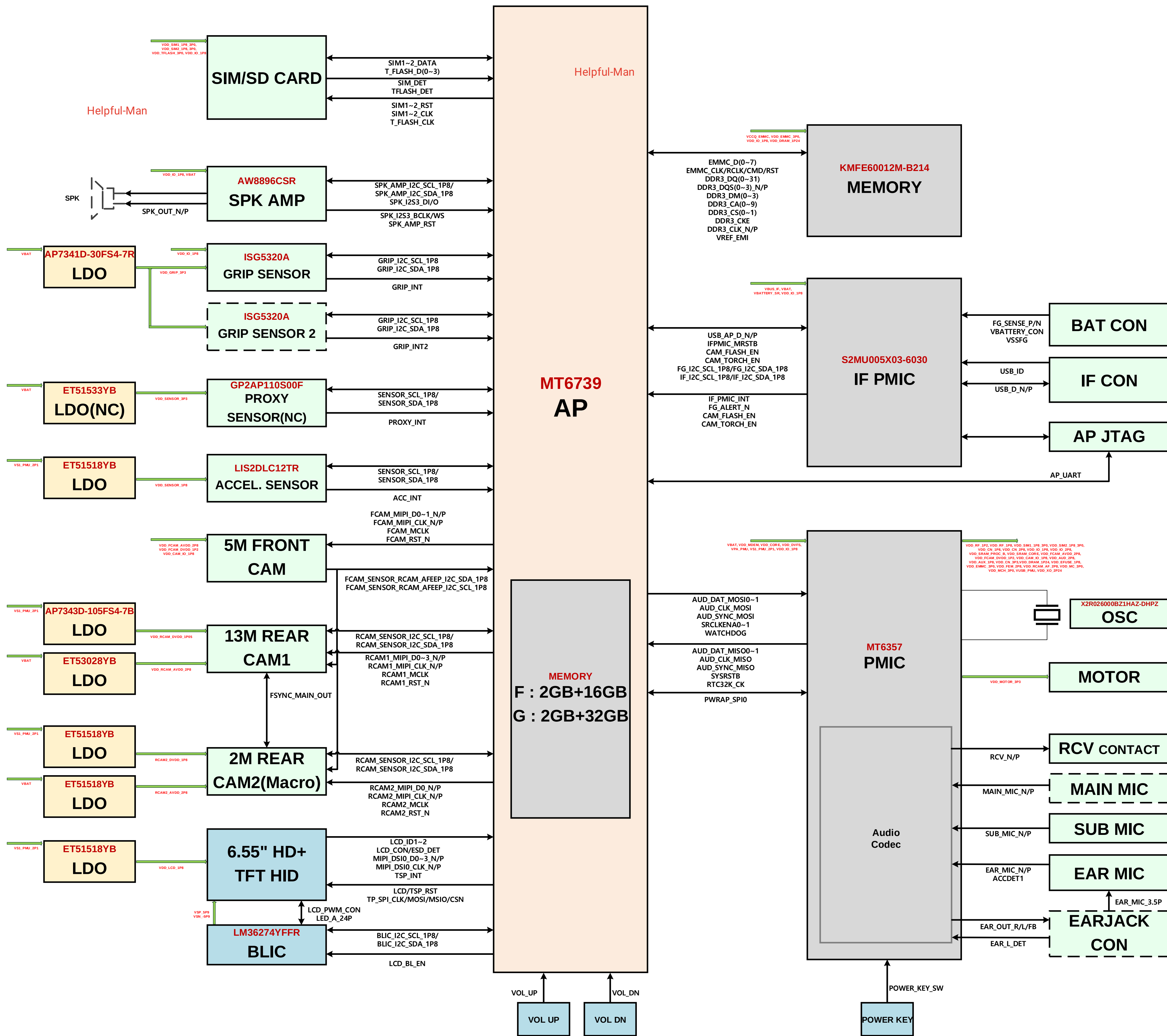
8. Level 3 Repair

SUB PBA (TOP SIDE)



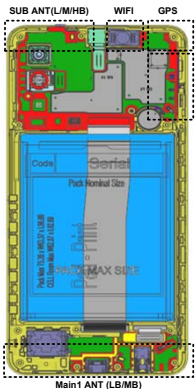
SUB PBA (BOTTOM SIDE)





SM-A022F/DS_REV1.0 RF Block Diagram

SM-A022F/G/M, M022F/G/M

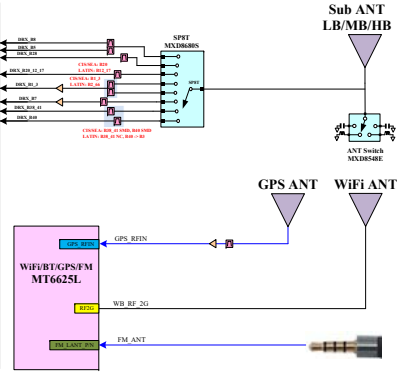
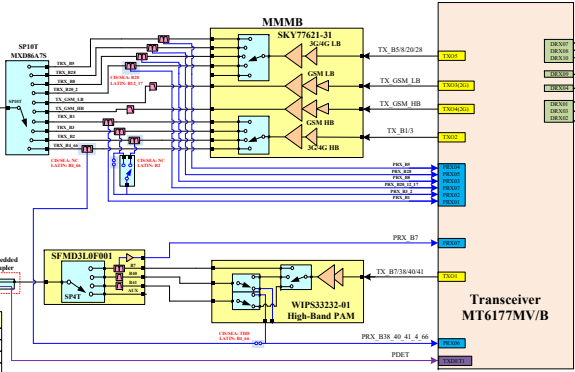


Main2 ANT (HB)

Main ANT LB/MB/HB

Main2 ANT HB

	CIS	SEA	LA
FDD	1/ 3/ 5/ 7/ 8/ 20	1/ 3/ 5/ 7/ 8/ 20/ 28	1/ 2/ 3/ 4/ 5/ 7/ 8/ 12/ 17/ 20/ 28/ 66
TDD	38/ 40/ 41	38/ 40/ 41	
3G	1/ 5/ 8	1/ 5/ 8	1/ 2/ 4/ 5/ 8
2G	850/ 900/ 1800	850/ 900/ 1800	850/ 900/ 1800/ 1900



SM-A022F/G/M, M022F/G/M
Common

SIM5001

SIM5002

SIM5000

SIM5003

Helpful-Man

SC6000

UCP400

UME5000

SOC7000

BOTH5000

CON5000

PM1

PM2

PM3

PM4

PM5

PM6

PM7

PM8

PM9

PM10

PM11

PM12

PM13

PM14

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PM320

PM321

PM322

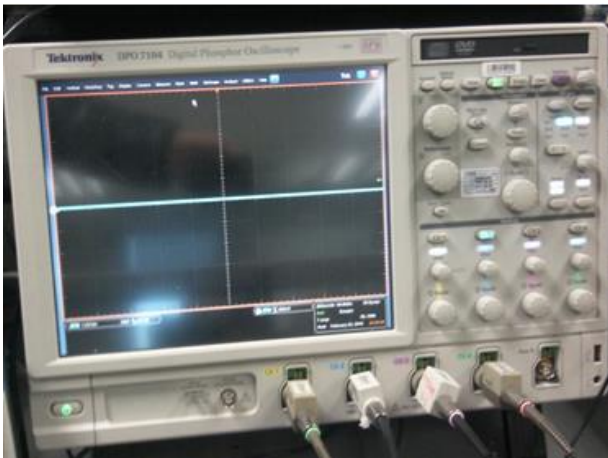
PM323

PM324

PM325

8. Level 3 Repair

8-3. Flow chart of Troubleshooting.



Oscilloscope



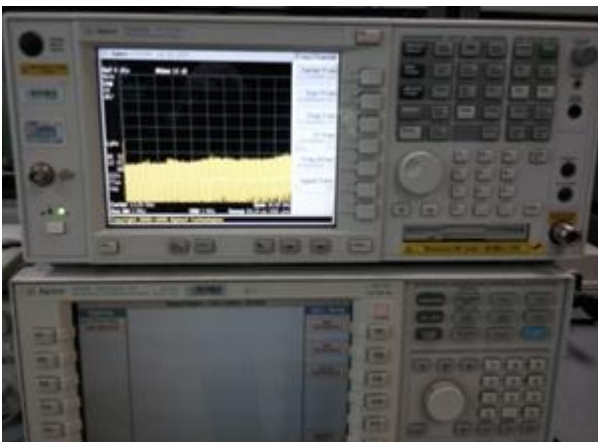
Digital Multimeter



Power Supply



+ driver, ESD Safe Tweezer



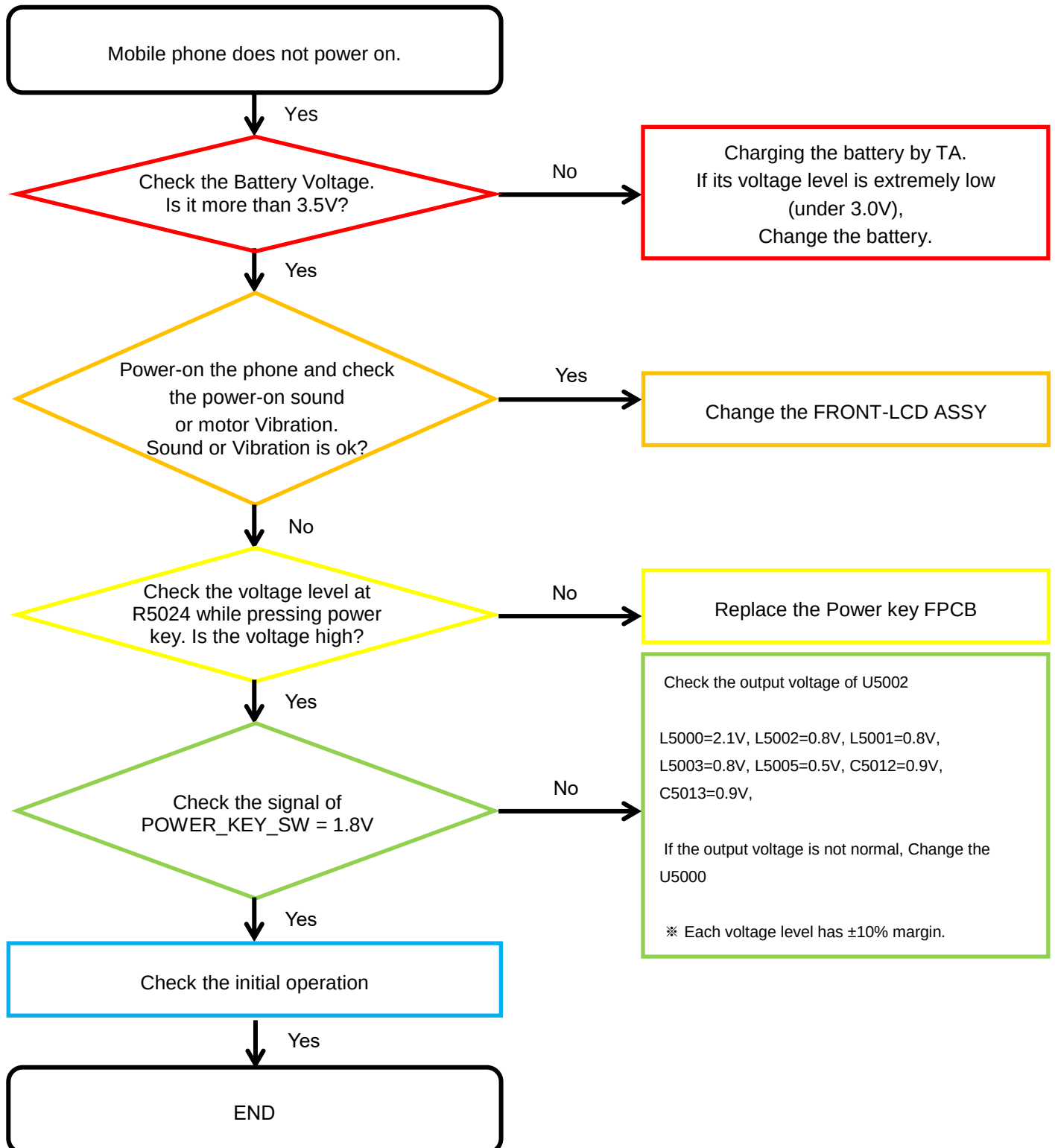
9960 & Spectrum Analyzer



Soldering iron

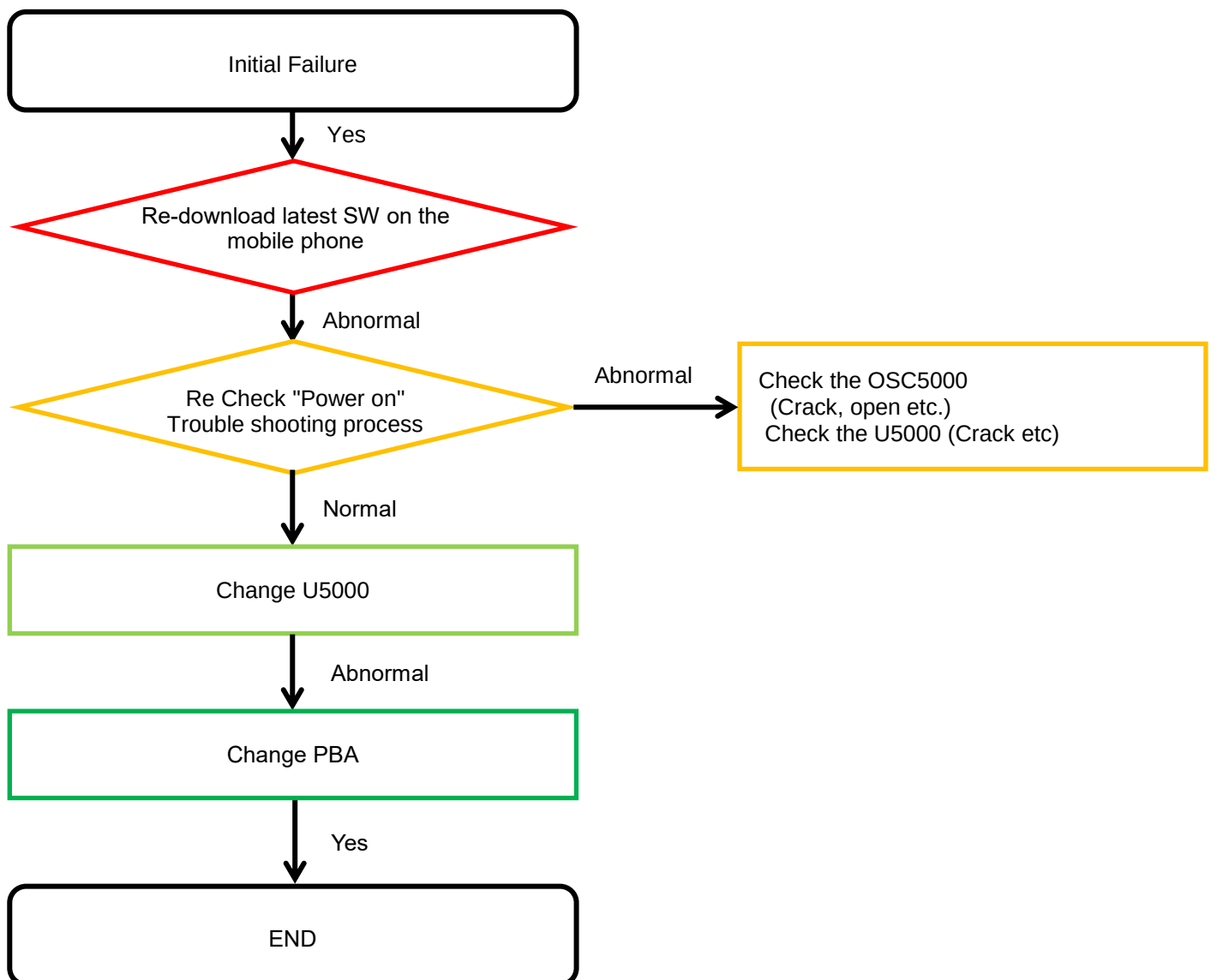
8. Level 3 Repair

8-4-1. Power On



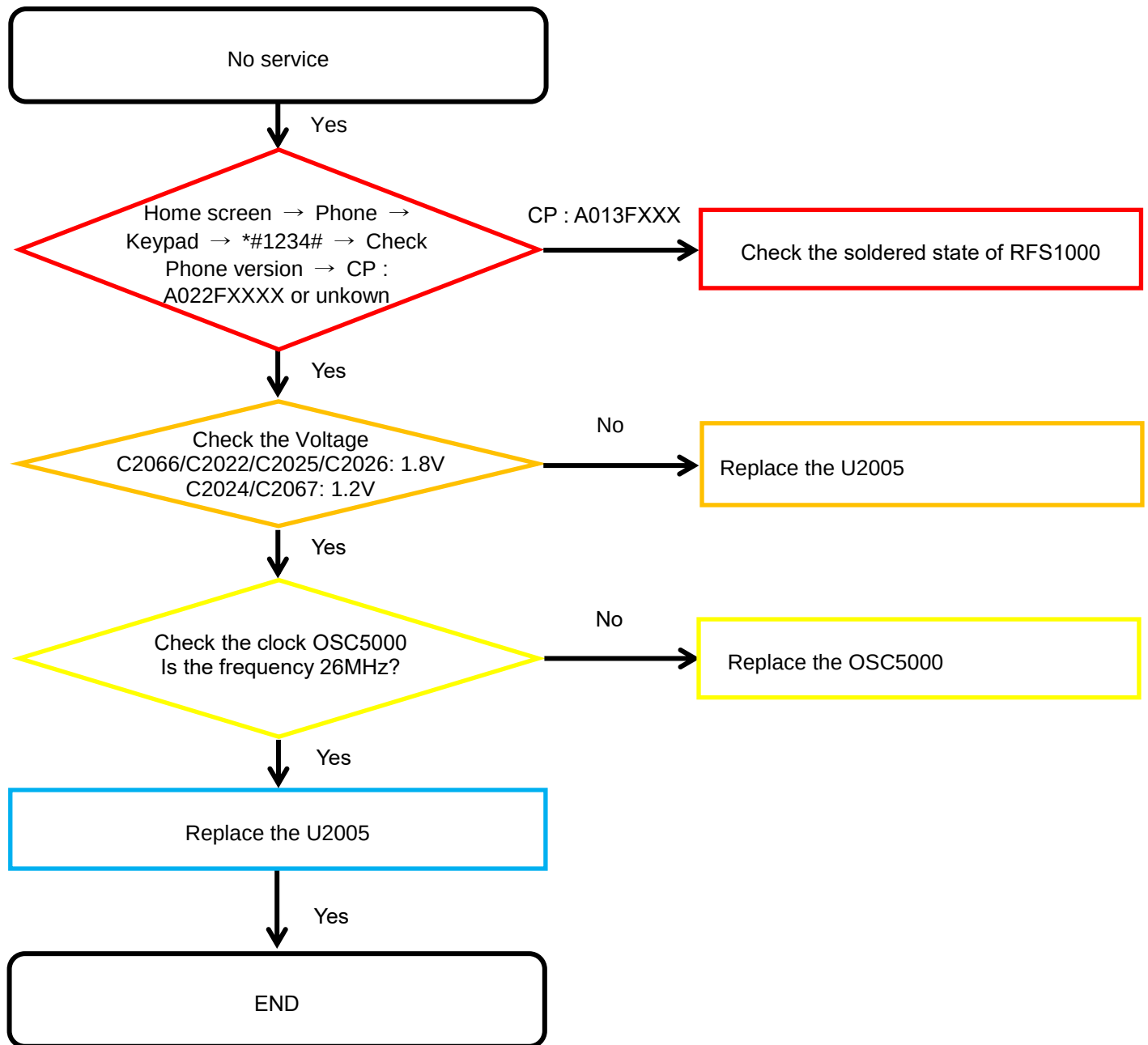
8. Level 3 Repair

8-4-2. Initial



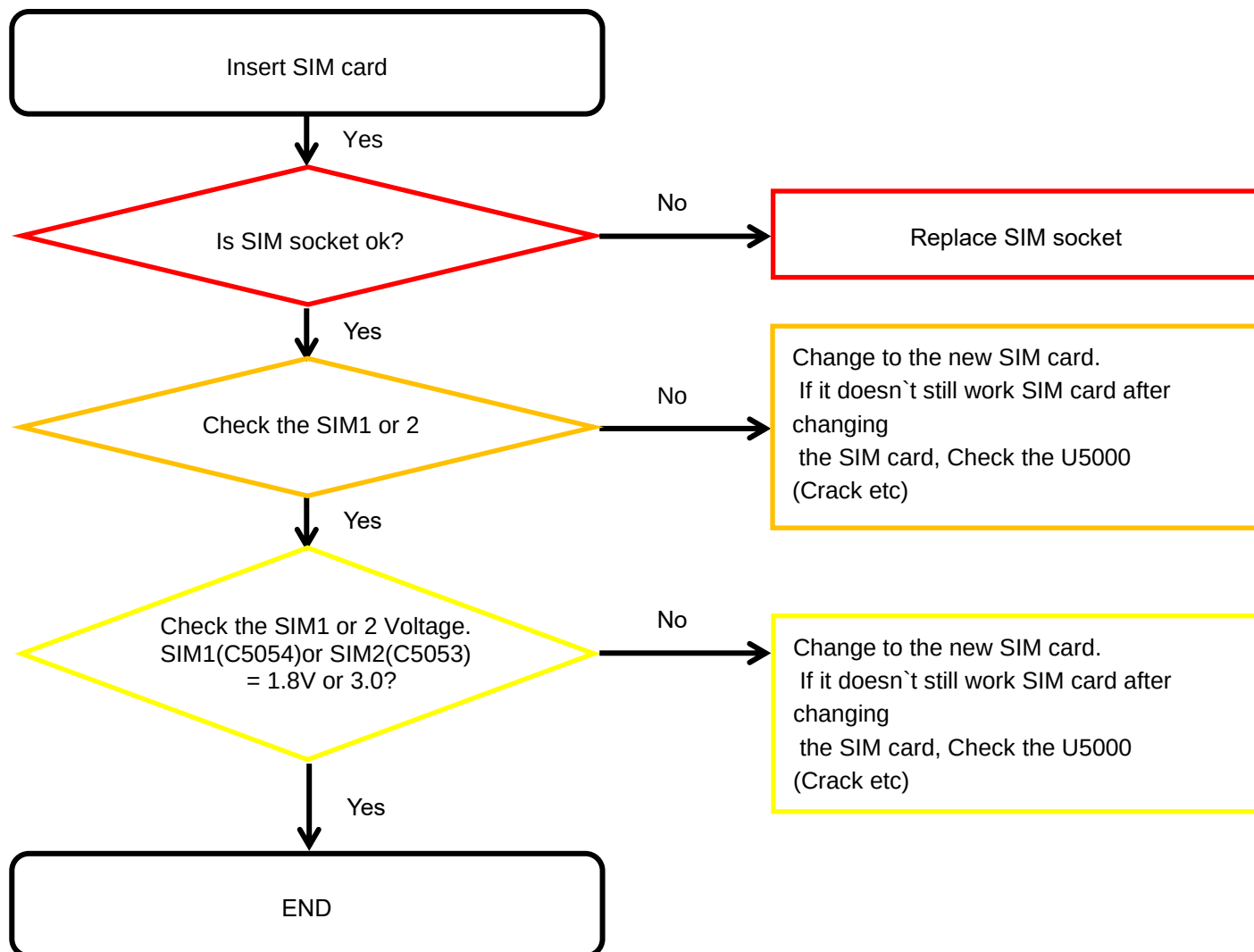
8. Level 3 Repair

8-4-3. No Service



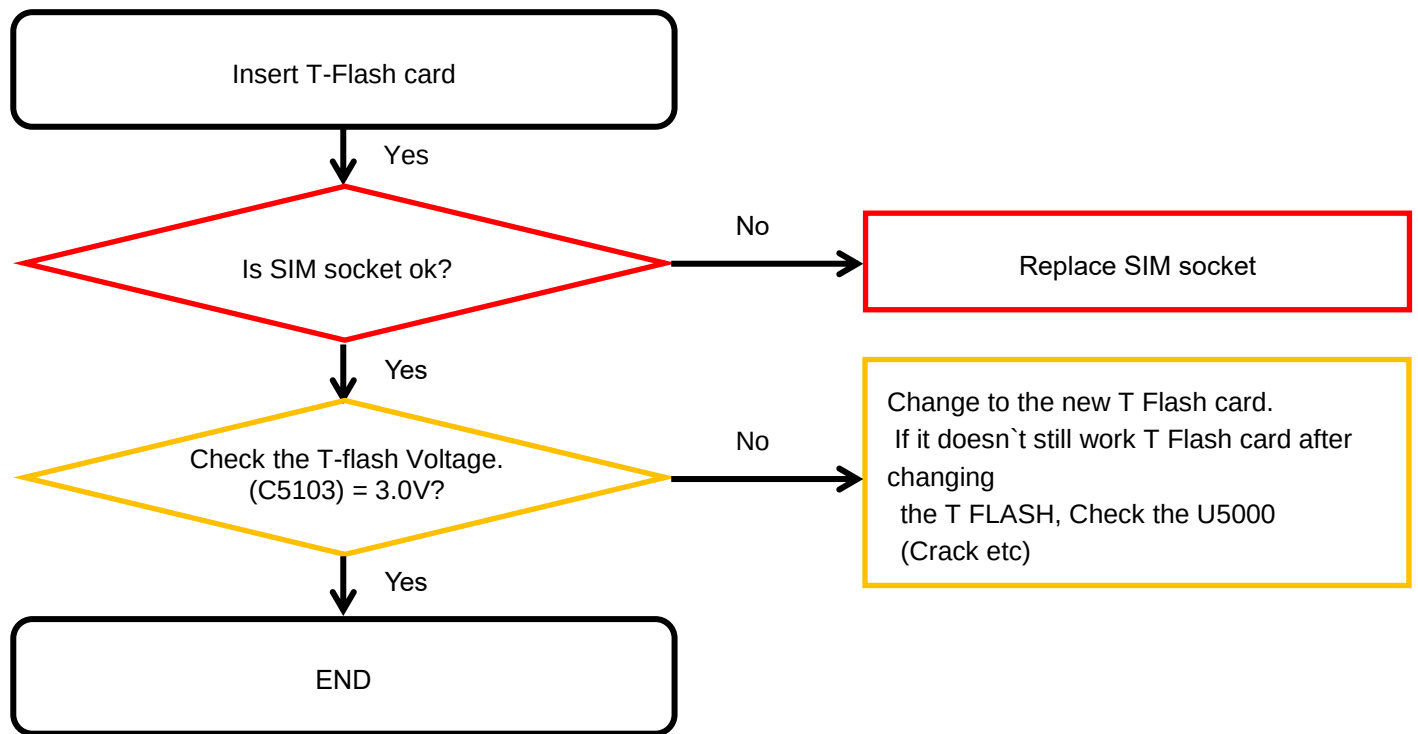
8. Level 3 Repair

8-4-4. SIM Part



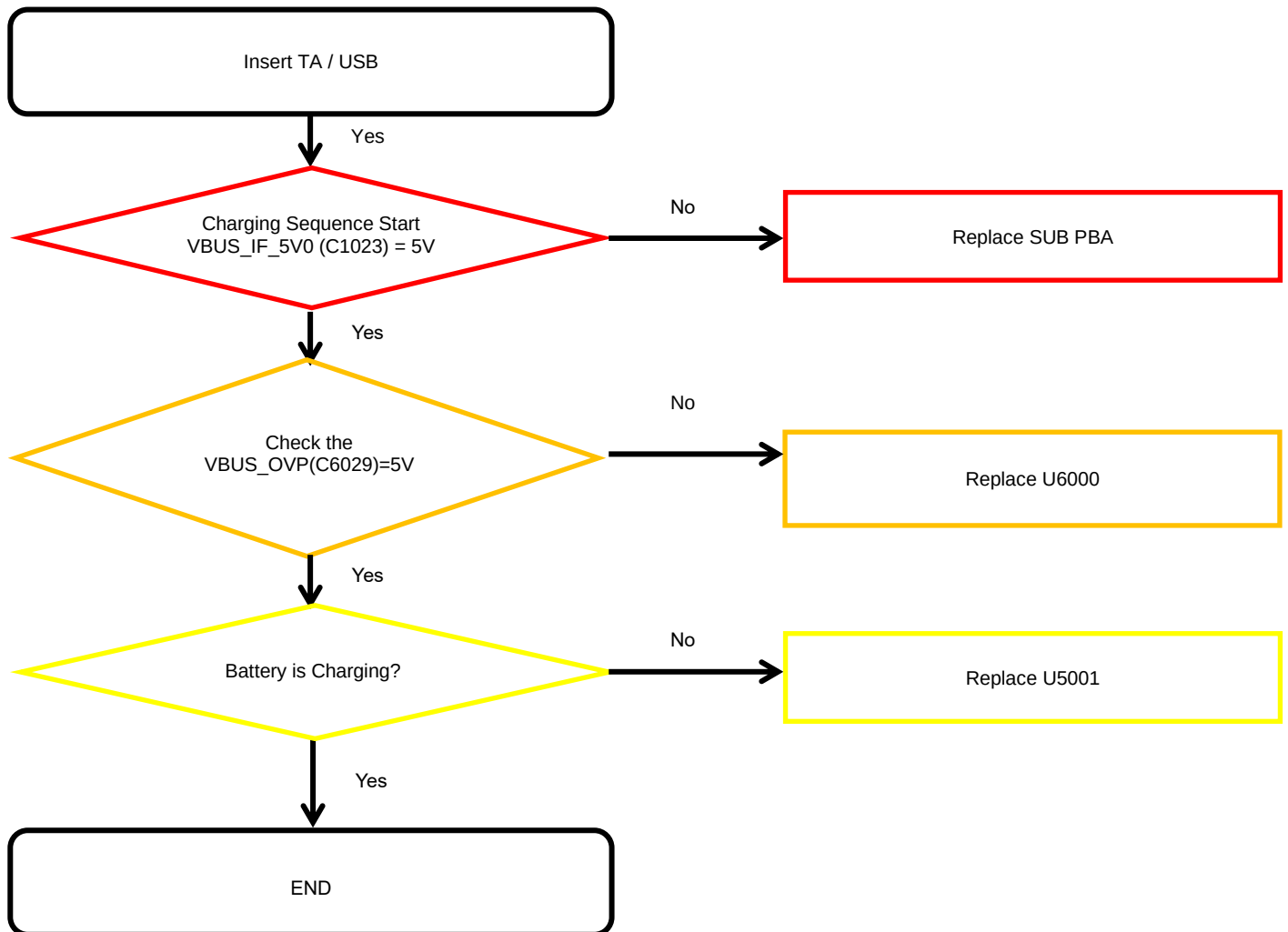
8. Level 3 Repair

8-4-5. T-Flash Part



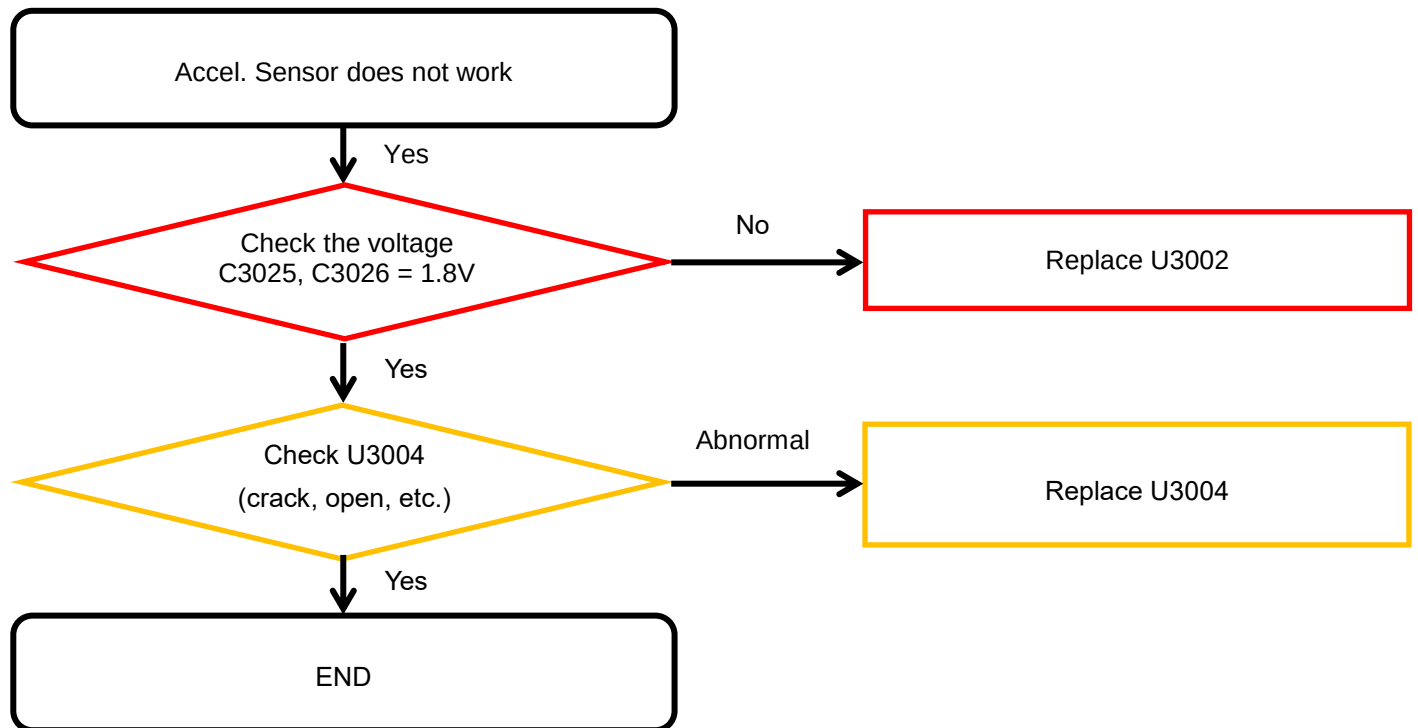
8. Level 3 Repair

8-4-6. Cable Charging Part



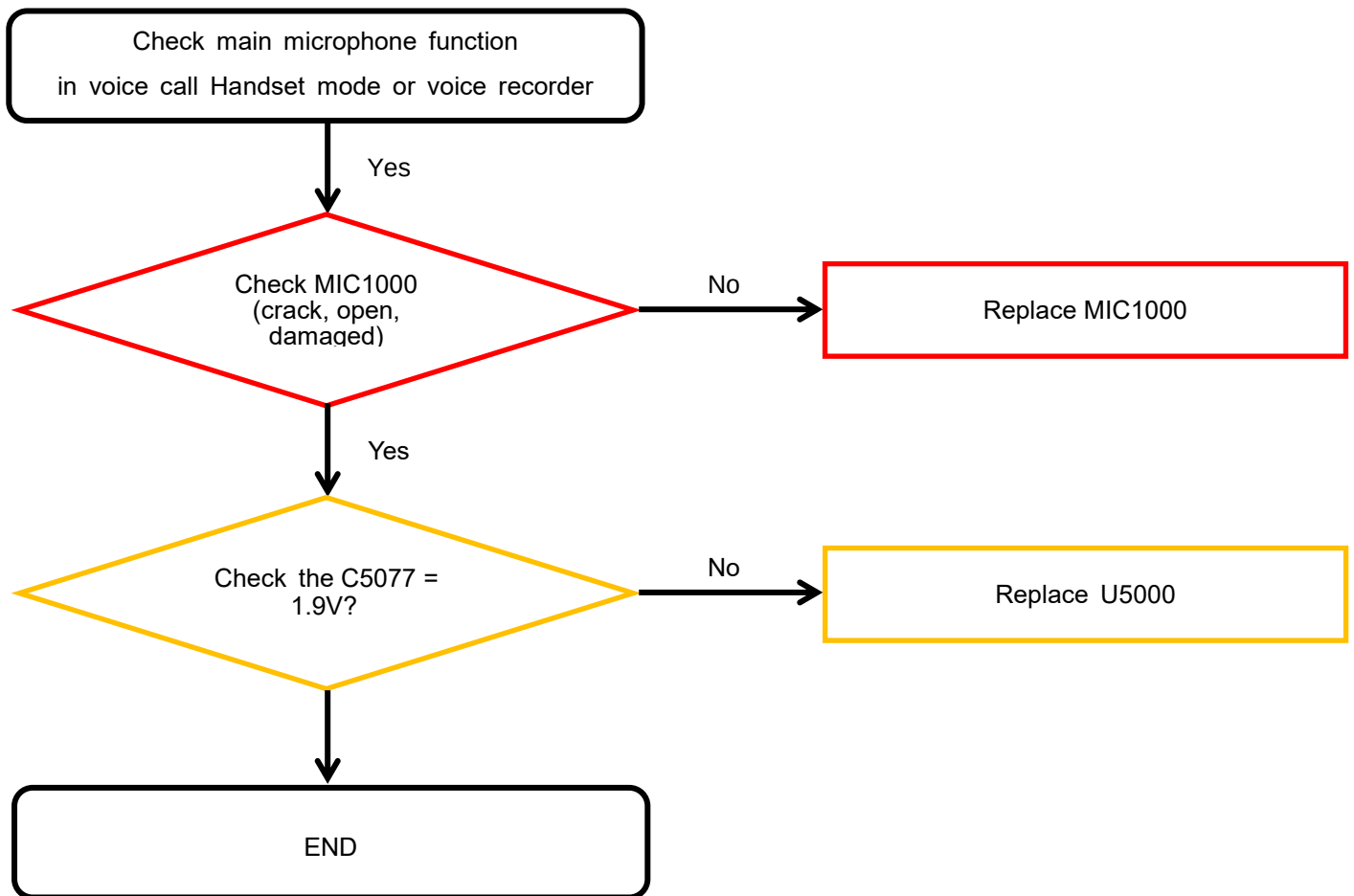
8. Level 3 Repair

8-4-7. Accel. Sensor



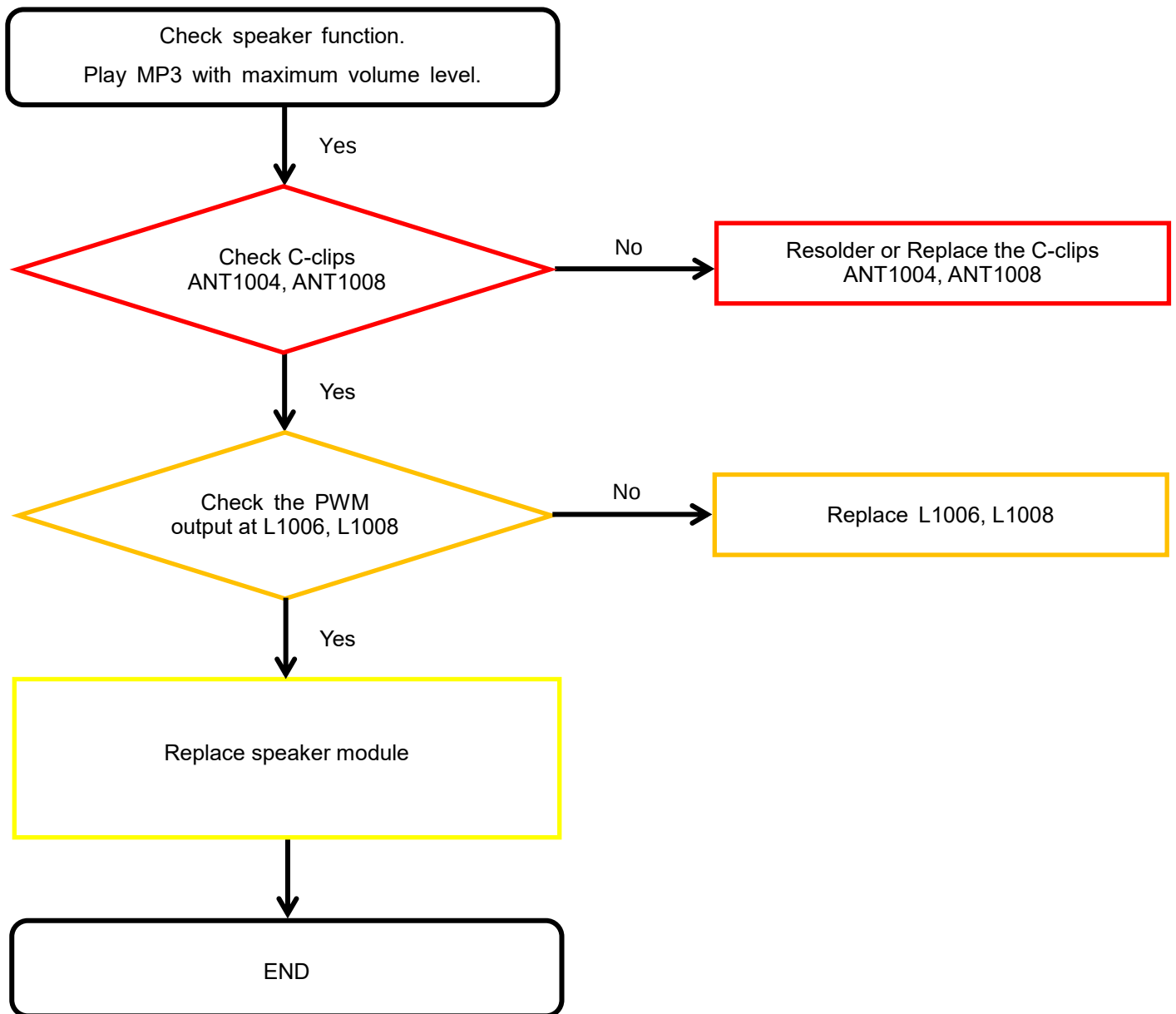
8. Level 3 Repair

8-4-8. Microphone Part - Main MIC



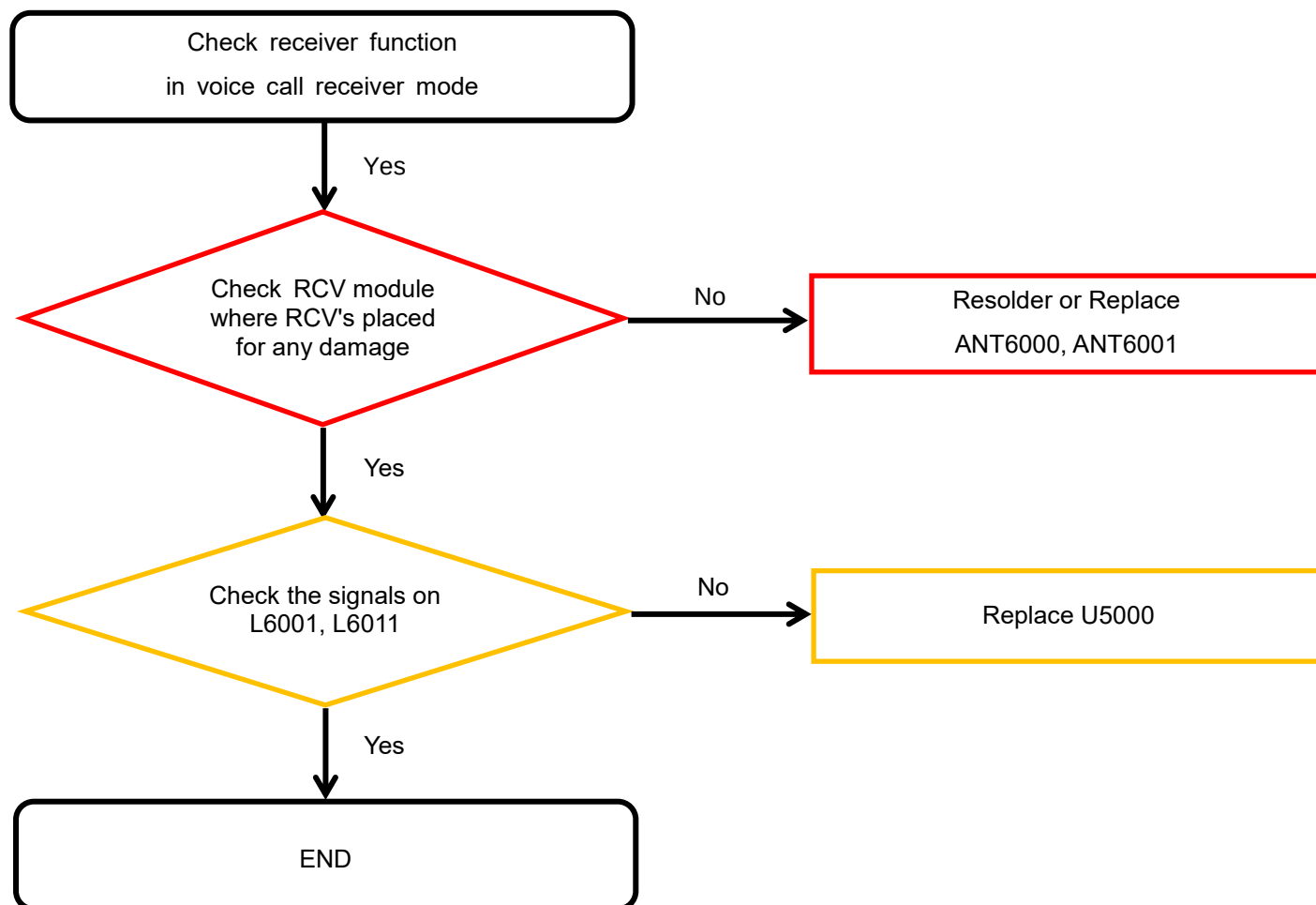
8. Level 3 Repair

8-4-9. Speaker Part



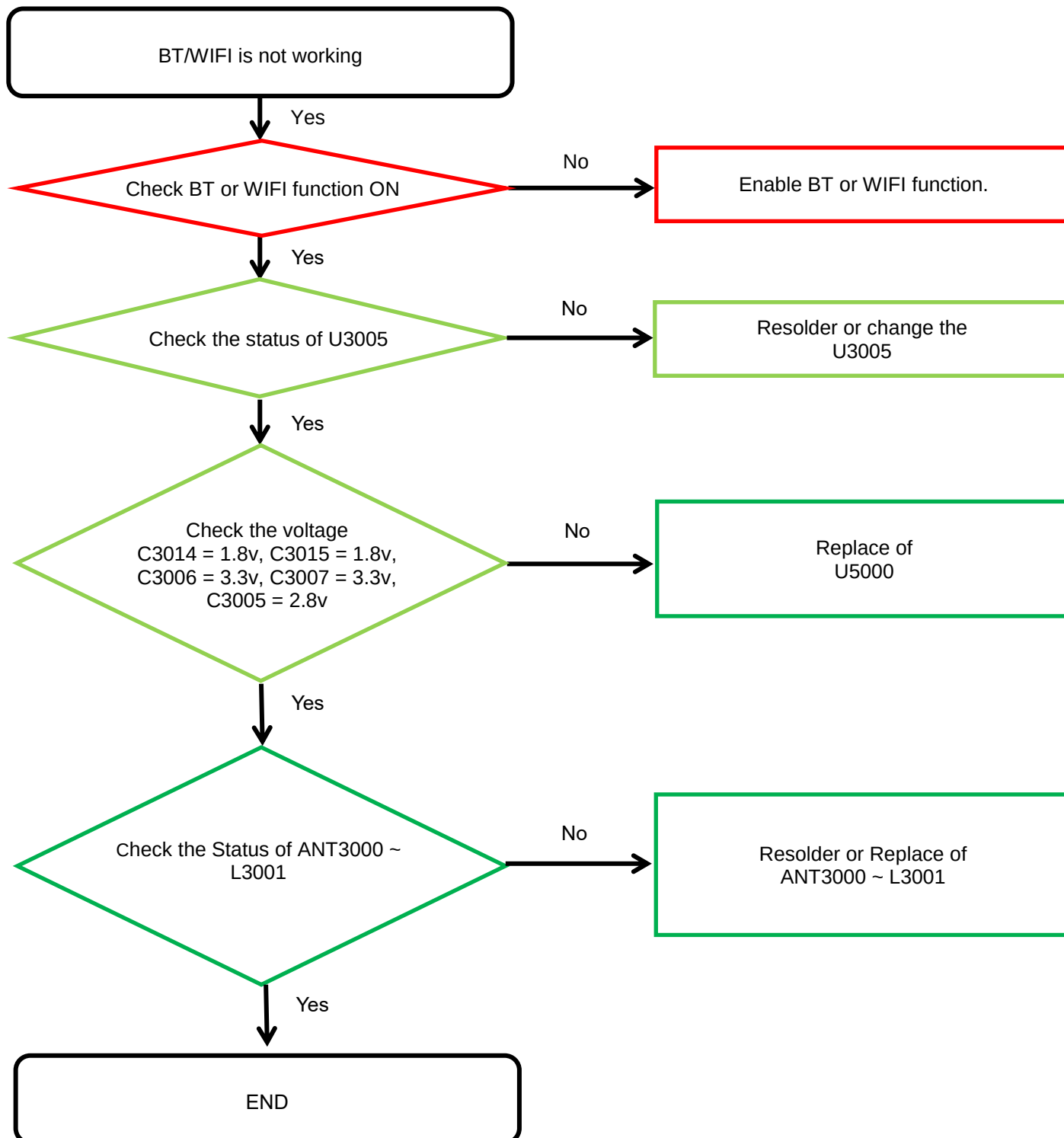
8. Level 3 Repair

8-4-10. Receiver Part



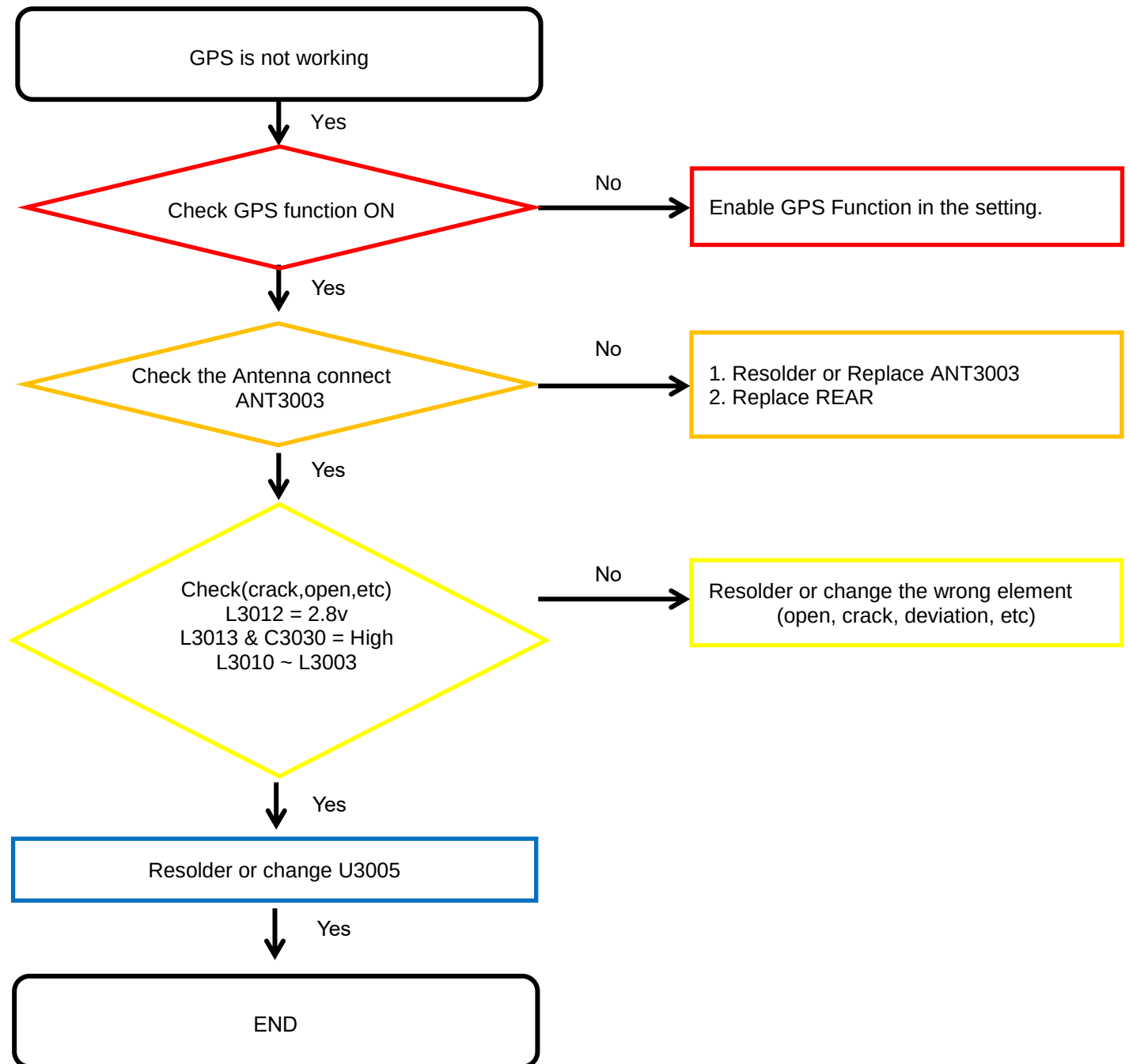
8. Level 3 Repair

8-4-11. BT/WIFI



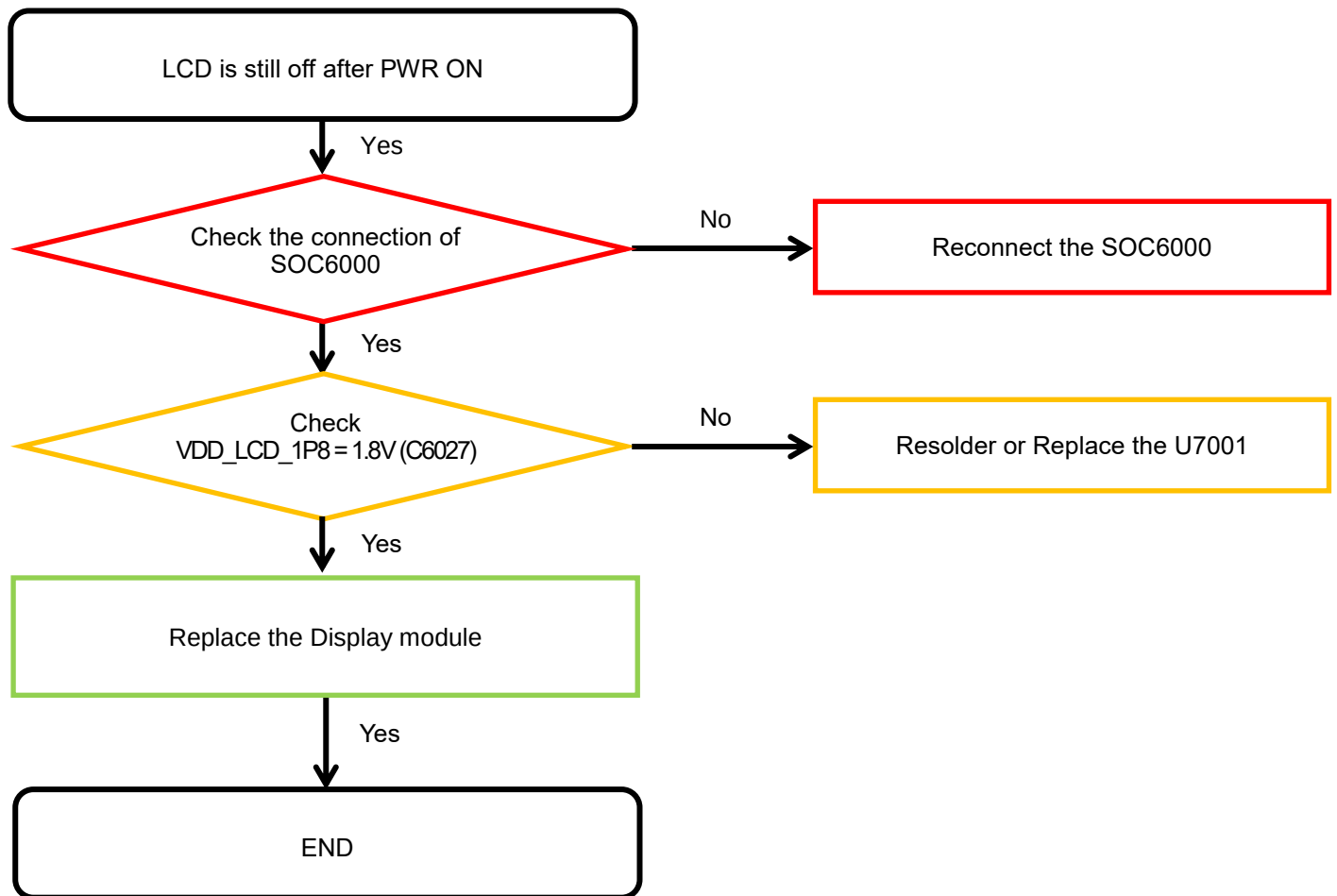
8. Level 3 Repair

8-4-12. GPS



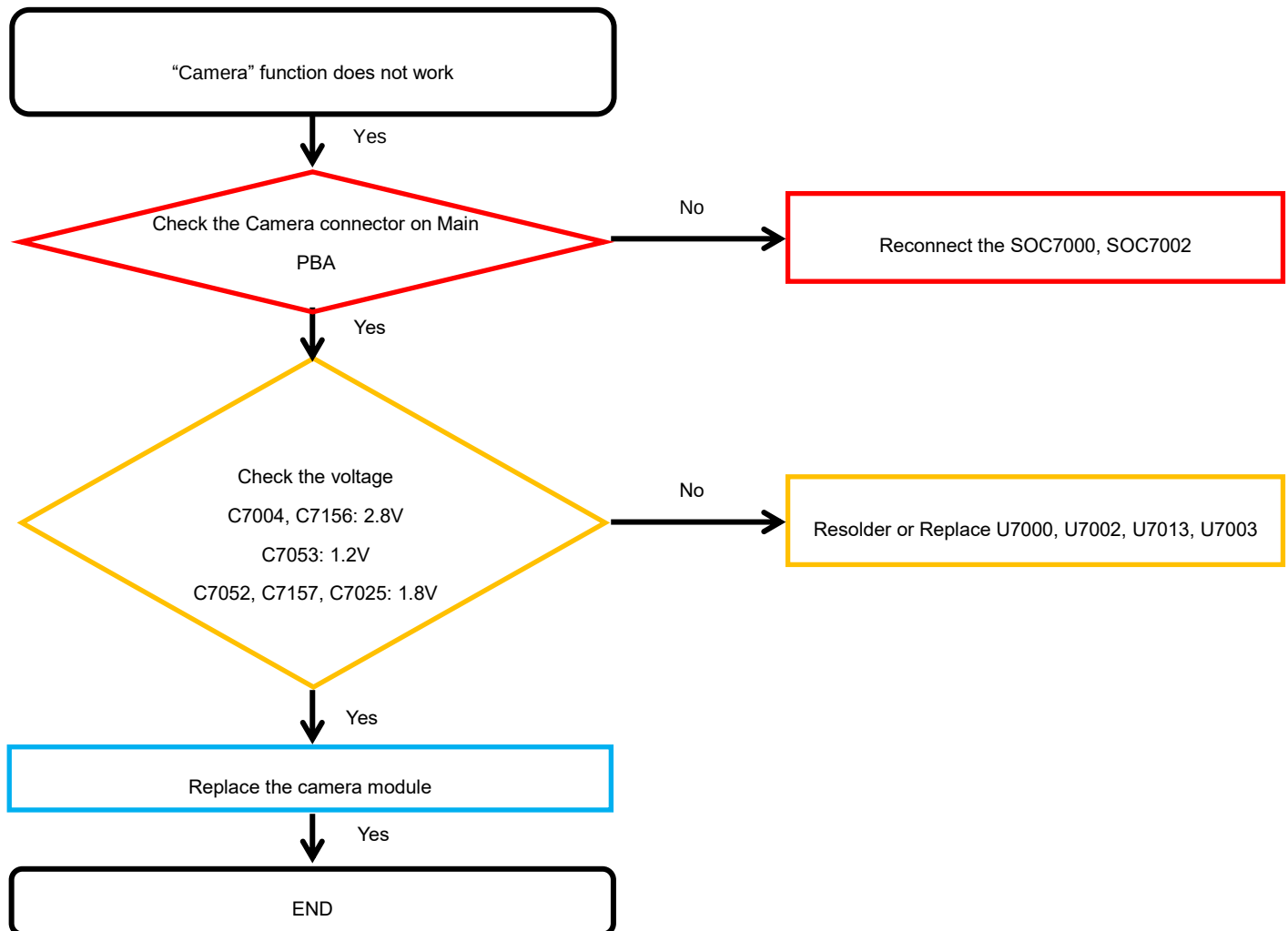
8. Level 3 Repair

8-4-13. Display



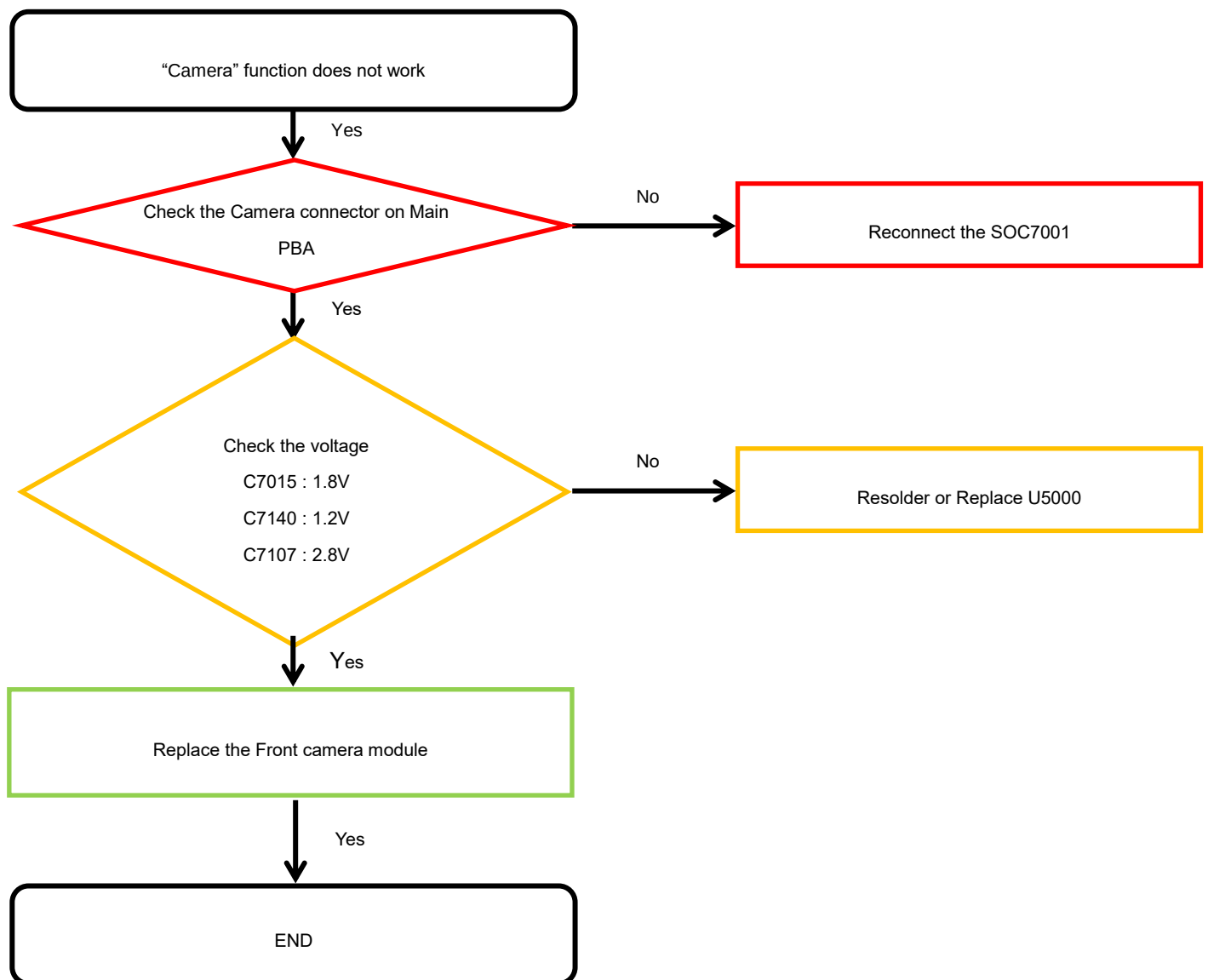
8. Level 3 Repair

8-4-14. 13M/2M REAR CAMERA



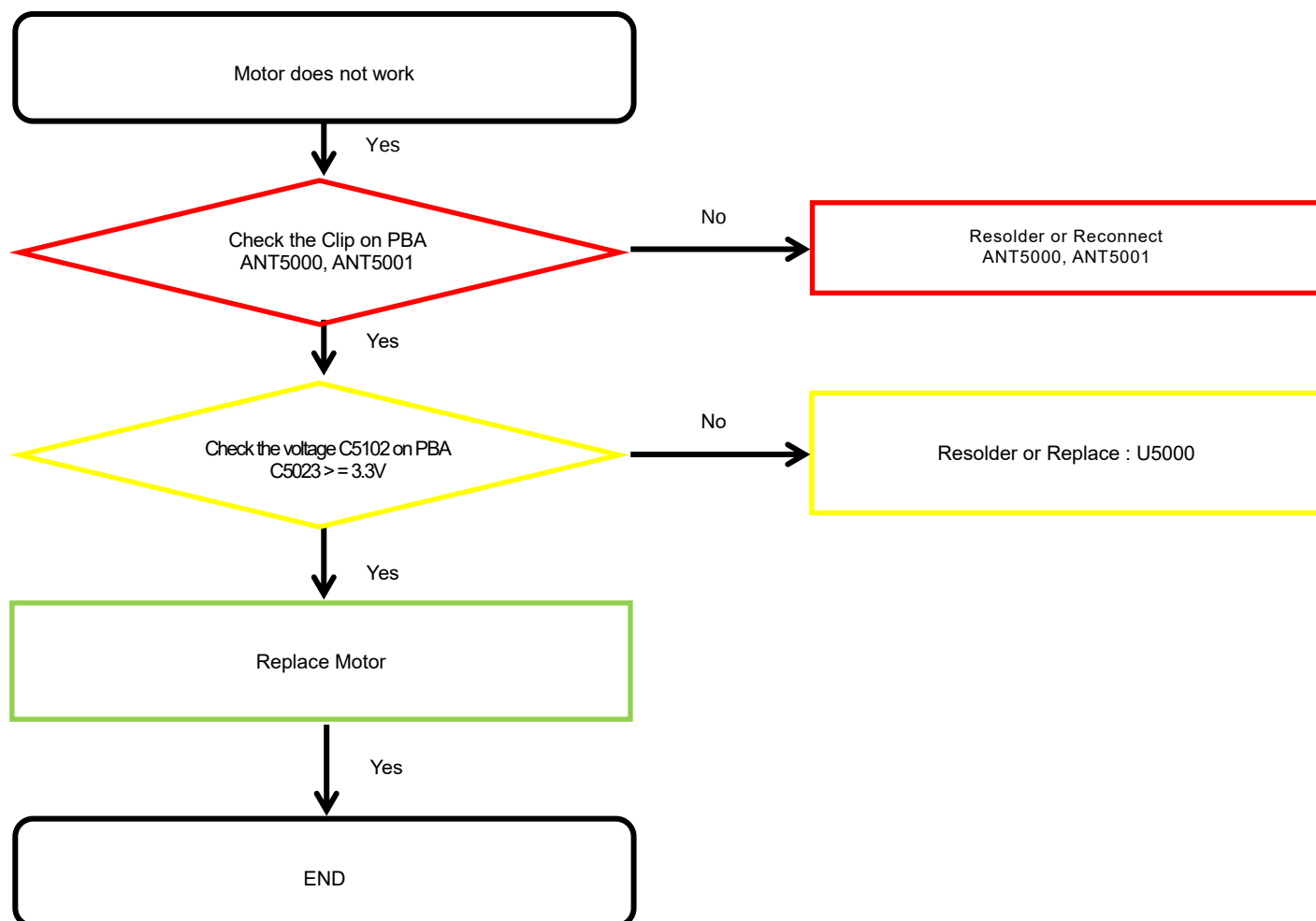
8. Level 3 Repair

8-4-15. 5M FRONT CAMERA



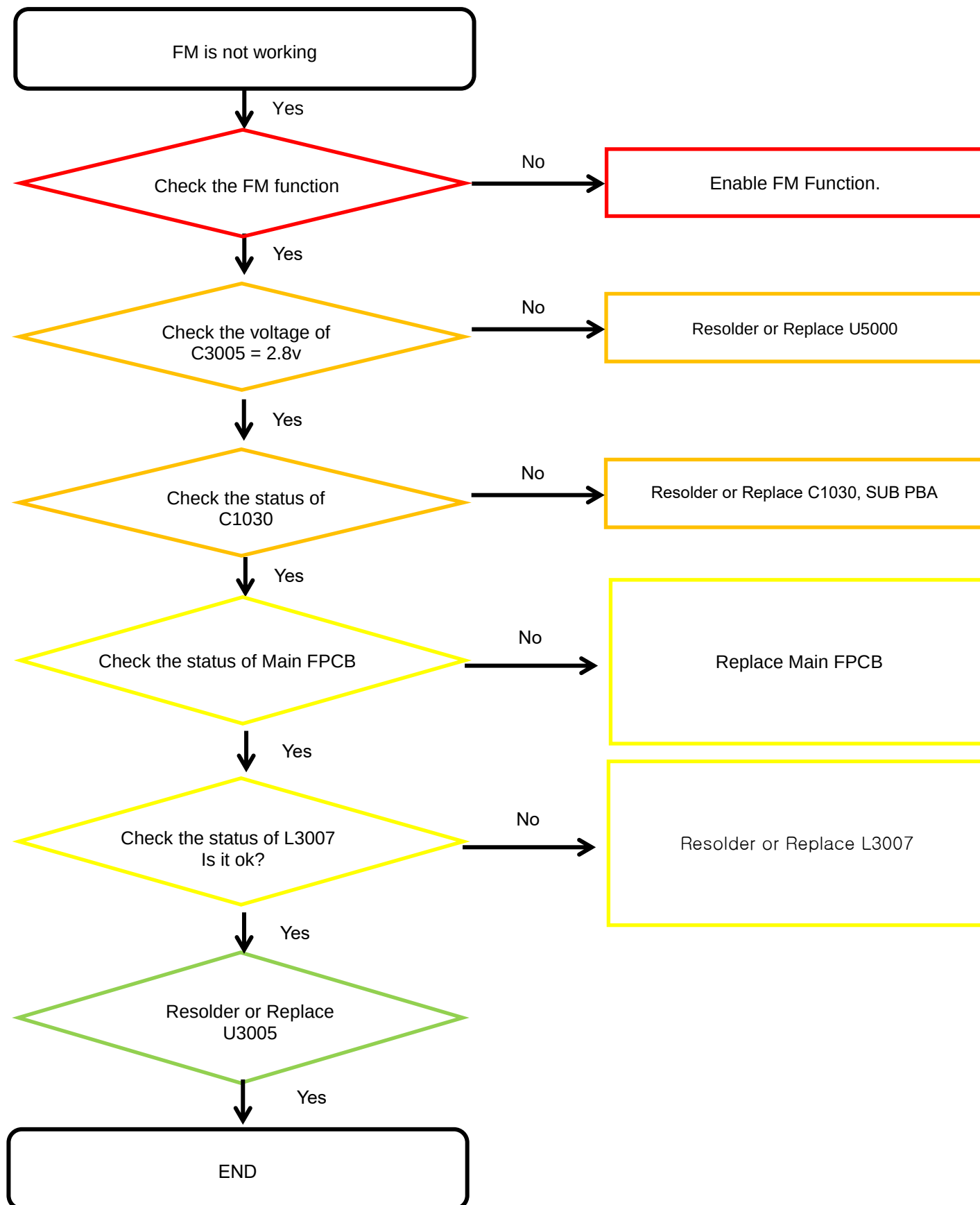
8. Level 3 Repair

8-4-16. Motor



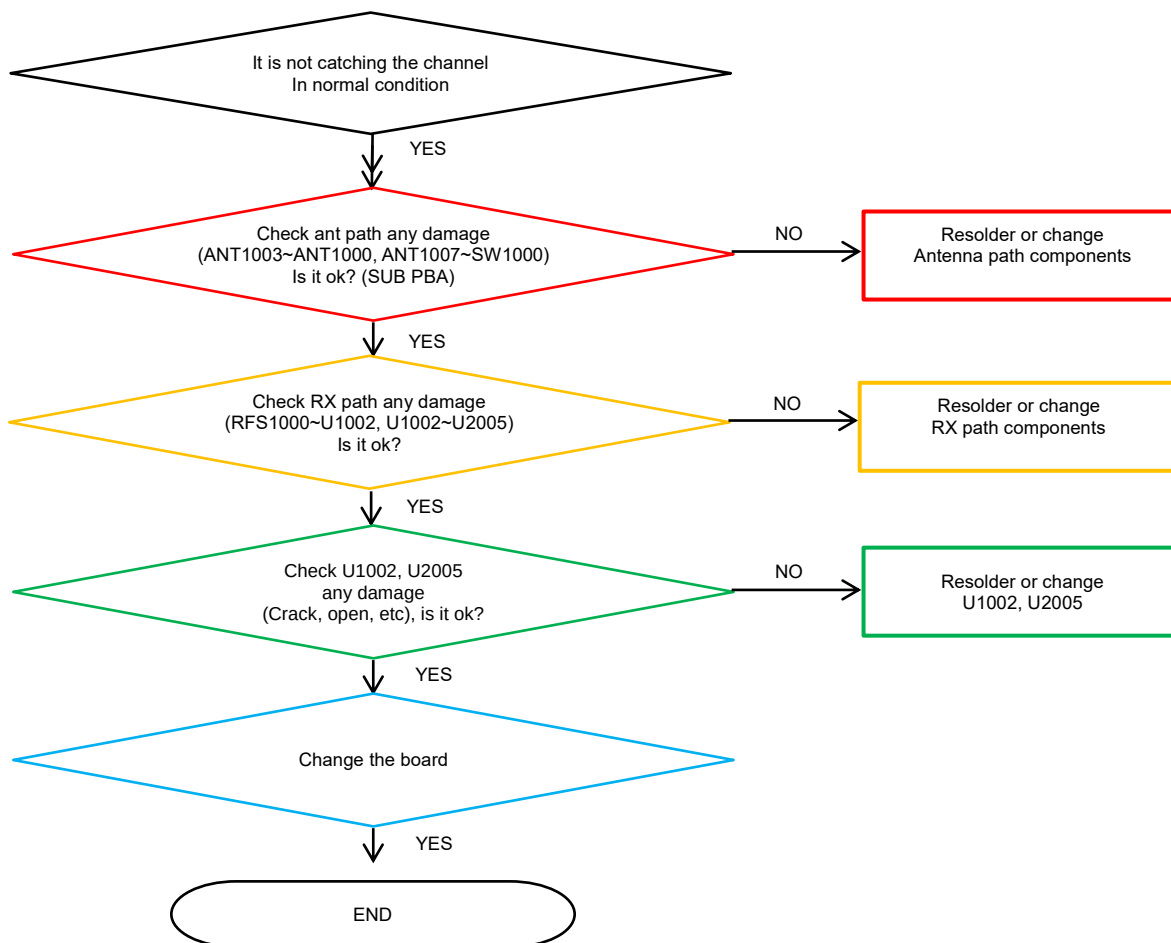
8. Level 3 Repair

8-4-17. FM radio part



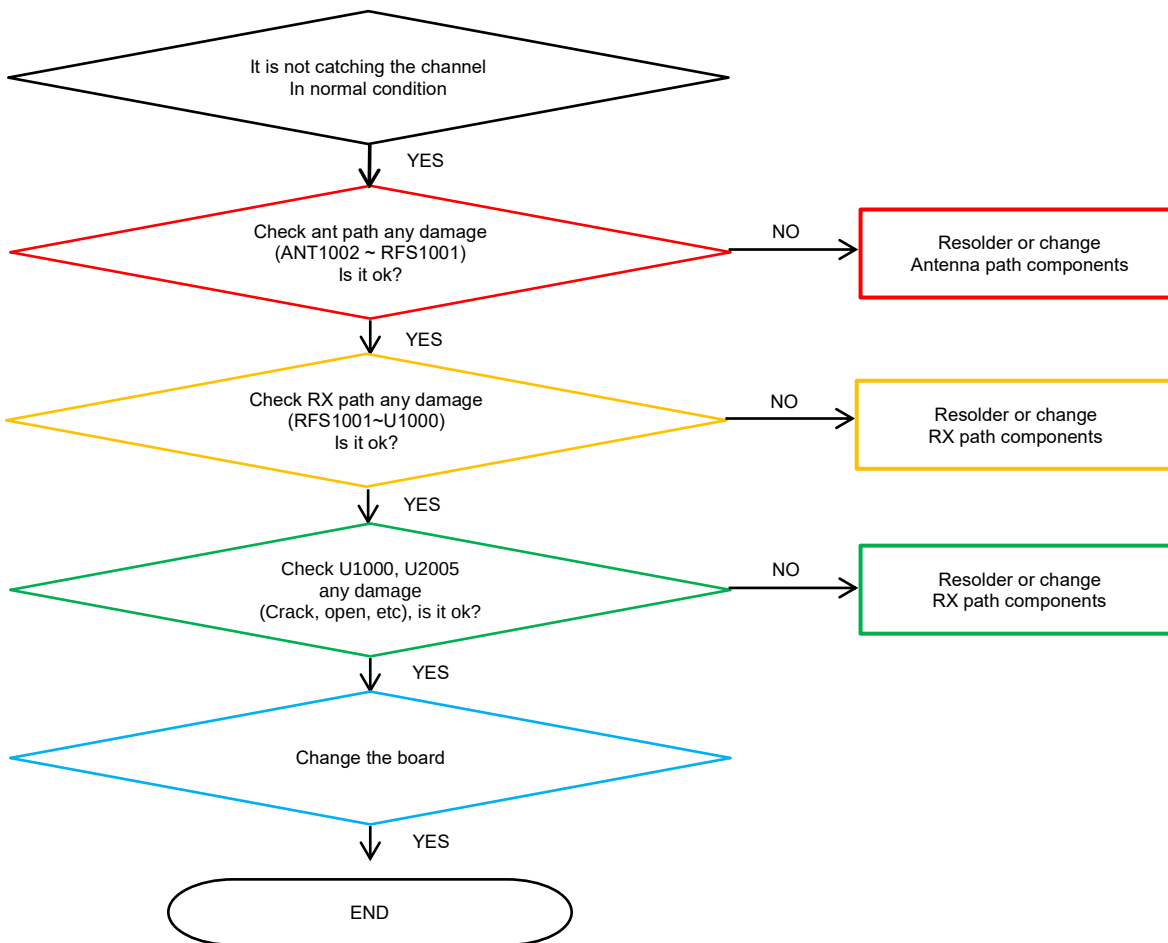
8. Level 3 Repair

8-4-18. GSM 850/900/1800/1900, WCDMA B1/2/4/5/8, LTE B1/2/3/4/5/8/12/17/20/28/66 PRx



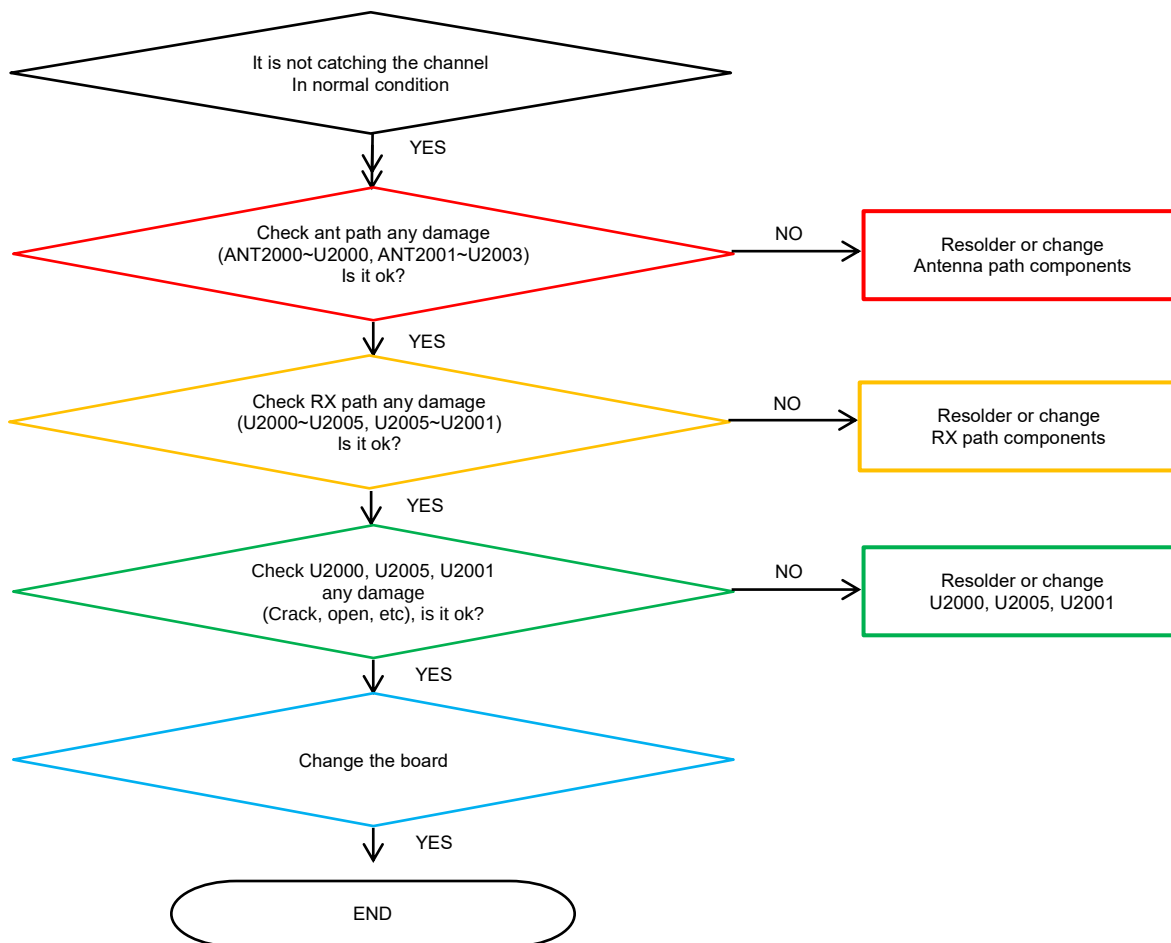
8. Level 3 Repair

8-4-19. LTE B7/38/40/41 PRx



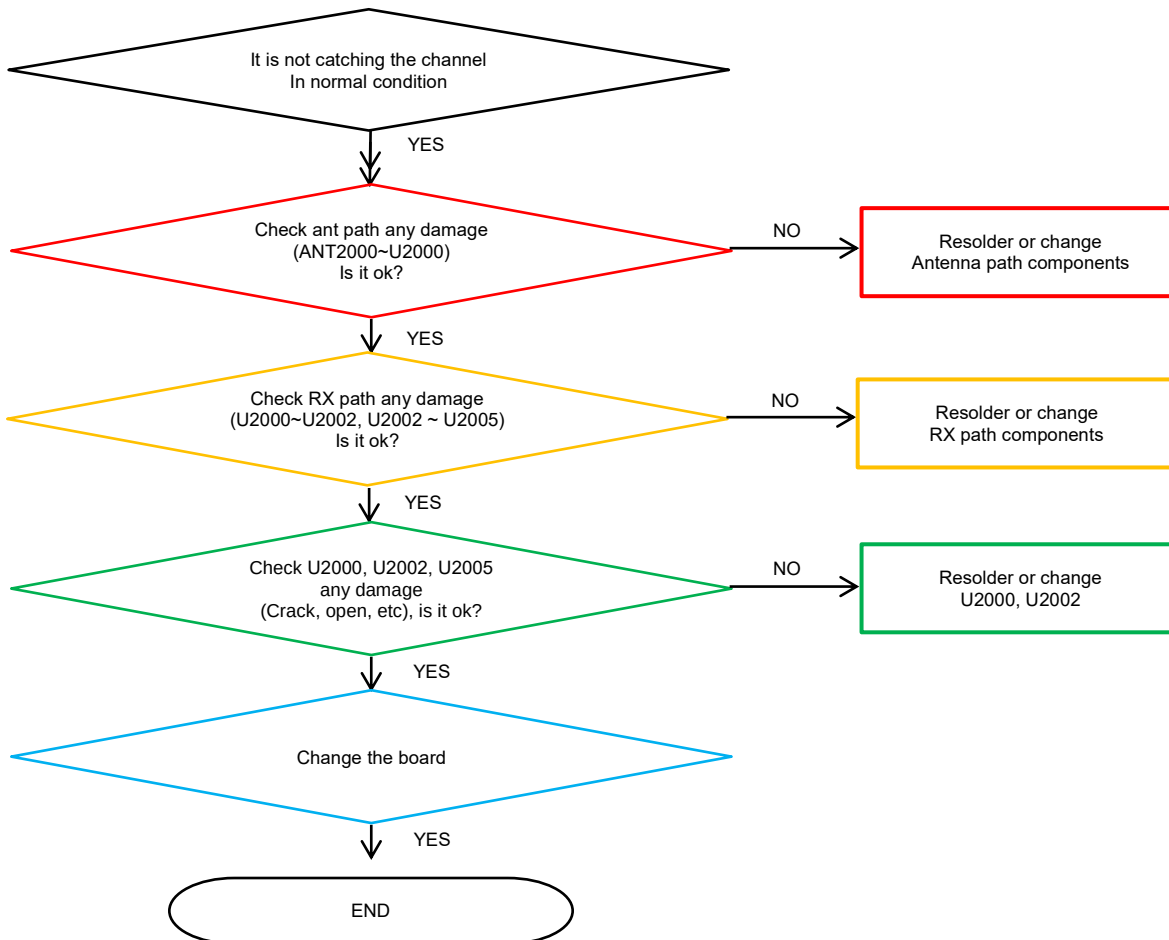
8. Level 3 Repair

8-4-20. WCDMA B1/2/4/5/8, LTE B1/2/3/4/5/8/12/17/20/28/66 DRx



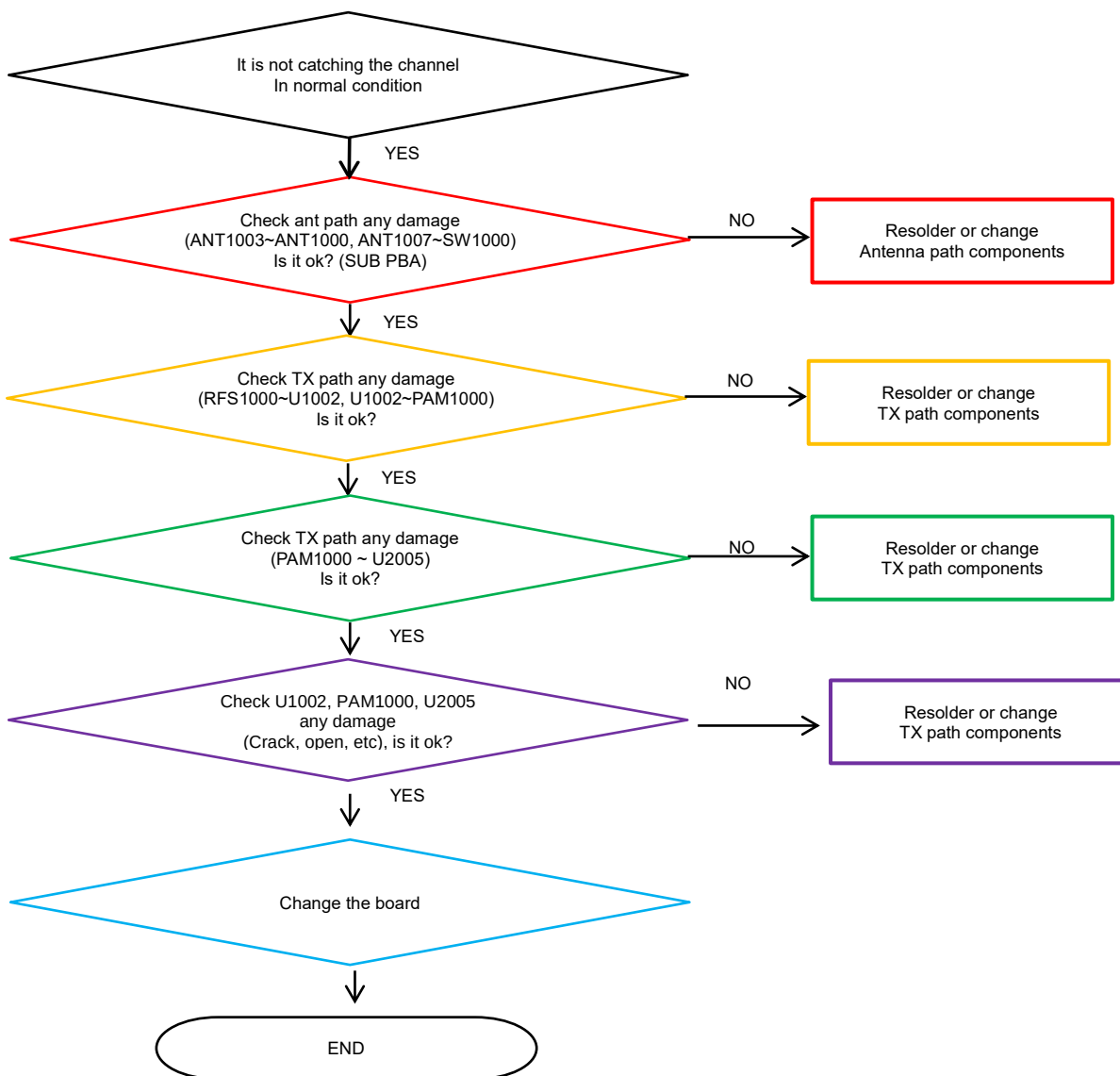
8. Level 3 Repair

8-4-21. LTE B7/38/40/41 DRx



8. Level 3 Repair

8-4-22. GSM 850/900/1800/1900, WCDMA B1/2/4/5/8, LTE B1/2/3/4/5/8/12/17/20/28/66 Tx



8. Level 3 Repair

8-4-23. LTE B7/38/40/41 Tx

